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Aircraft restroom unit

Abstract

A structural frame of an aircraft lavatory unit includes a first side wall facing a latitudinal aisle, and boarding entrances are located at both ends of the latitudinal aisle. A pump-type bottle that can discharge a disinfectant for hands and fingers is disposed at the first side wall at a height at which a passenger can push down a head portion thereof.

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Background/Summary

TECHNICAL FIELD

(1) The present technology relates to an aircraft lavatory unit.

BACKGROUND ART

(2) In recent years, measures for preventing infections with a novel coronavirus have been widely taken in various countries around the world. Most of the infections with the novel coronavirus are said to be caused by touching eyes, a nose, or a mouth with hands and fingers with viruses attached due to contact with objects.

(3) It is preferable to take measures to prevent the infections with the novel coronavirus also in an aircraft cabin.

(4) The present inventor intensively studies prevention of spread of passenger-to-passenger infections with the novel coronavirus attached to hands and fingers due to contact with objects in an aircraft cabin, and focuses on a handle of a door for opening and closing an entrance of an aircraft lavatory unit.

(5) That is, on a long-distance flight for five or six hours or more, every passenger uses the aircraft lavatory unit at least once.

(6) In other words, on the long-distance flight, every passenger touches the handle of the entrance door of the aircraft lavatory unit.

(7) Then, if cabin attendants disinfect the handle of the entrance door of the aircraft lavatory unit every time a passenger uses the aircraft lavatory unit, there is no possibility of infecting other passengers even with the novel coronavirus that has attached to the handle.

(8) Unfortunately, some airlines reduce the number of cabin attendants from the viewpoint of cost reduction or the like, and thus cabin attendants may not be able to disinfect the handle of the entrance door of the aircraft lavatory unit every time a passenger uses the aircraft lavatory unit.

(9) In such a case, passengers who do not want to be infected with the novel coronavirus can disinfect their hands and fingers with, for example, own tissues impregnated with a disinfectant after leaving the aircraft lavatory unit but cannot disinfect their hands and fingers if they forget to bring tissues impregnated with a disinfectant.

SUMMARY

(10) The present technology provides an aircraft lavatory unit that enables passengers to disinfect their hands and fingers every time they use the aircraft lavatory unit and is advantageous in preventing infections with a novel coronavirus in a cabin.

(11) One embodiment of the present technology is an aircraft lavatory unit disposed in an aircraft comprising a longitudinal aisle extending in a front-rear direction of a fuselage, a latitudinal aisle extending perpendicularly to the longitudinal aisle, and seat groups provided on both sides of the longitudinal aisle, the aircraft lavatory unit facing at least one of the longitudinal aisle or the latitudinal aisle. A structural frame of the aircraft lavatory unit includes a wall portion facing the at

least one of the longitudinal aisle or the latitudinal aisle, and a pump-type bottle that can discharge a disinfectant for hands and fingers is disposed at the wall portion at a height at which a passenger can push down a head portion of the pump-type bottle.

(12) According to the present technology, when using the aircraft lavatory unit, passengers can push down the head portion of the pump-type bottle to discharge a disinfectant to their hands and fingers and easily disinfect them.

(13) Therefore, even if every passenger touches the handle of the aircraft lavatory unit on a long-distance flight, the embodiment is advantageous in preventing the infections with the novel coronavirus to other passengers via the handle of the aircraft lavatory unit and thus is advantageous in preventing the infections with the novel coronavirus in the aircraft cabin.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a perspective view of an aircraft lavatory unit to which a first embodiment is applied.

(2) FIG. 2 is a perspective view illustrating a pump-type bottle disposed at a first side wall by using an arrangement instrument according to the first embodiment, in which male thread members are omitted to clarify the structure of the arrangement instrument.

(3) FIG. 3 is a cross-sectional view illustrating the pump-type bottle disposed at the first side wall by using the arrangement instrument according to the first embodiment.

(4) FIG. 4A is a perspective view of a lower bracket according to the first embodiment, and FIG. 4B is a side view thereof.

(5) FIG. 5A is a perspective view of an upper bracket according to the first embodiment, and FIG. 5B is a front view thereof.

(6) FIG. 6 is a perspective view of an aircraft lavatory unit to which a second embodiment is applied.

(7) FIGS. 7A-7B include explanatory diagrams of a third embodiment, where FIG. 7A is a perspective view of a lower bracket according to the third embodiment, and FIG. 7B is a side view thereof.

(8) FIG. 8 is an explanatory diagram of a fourth embodiment and is a perspective view illustrating the pump-type bottle disposed at the first side wall by using an arrangement instrument according to the fourth embodiment, in which male thread members are omitted to clarify the structure of the arrangement instrument.

(9) FIG. 9 is a cross-sectional view illustrating the pump-type bottle disposed at the first side wall by using the arrangement instrument according to the fourth embodiment.

(10) FIG. 10A is a perspective view of a lower bracket according to the fourth embodiment, and FIG. 10B is a plan view thereof.

(11) FIGS. 11A-11B include explanatory diagrams of a fifth embodiment, where FIG. 11A is a perspective view of a lower bracket according to the fifth embodiment, and FIG. 11B is a plan view of the lower bracket in which a fourth bolt constituting a bottom portion of the lower bracket is attached to the first side wall.

(12) FIG. 12 is an explanatory diagram of a sixth embodiment, and is a perspective view illustrating the pump-type bottle disposed at the first side wall by using an arrangement instrument according to the sixth embodiment, in which male thread members are omitted to clarify the structure of the arrangement instrument.

(13) FIG. 13 is a cross-sectional view illustrating the pump-type bottle disposed at the first side wall by using the arrangement instrument according to the sixth embodiment.

(14) FIG. 14 is an explanatory diagram of a seventh embodiment and is a perspective view illustrating the pump-type bottle disposed at the first side wall by using an arrangement instrument

according to the seventh embodiment, in which male thread members are omitted to clarify the structure of the arrangement instrument.

(15) FIG. 15 is a cross-sectional view illustrating the pump-type bottle disposed at the first side wall by using the arrangement instrument according to the seventh embodiment.

(16) FIG. 16 is a plan view of a rear portion of an aircraft cabin.

DETAILED DESCRIPTION

(17) Next, an aircraft lavatory unit according to an embodiment of the present technology will be described with reference to drawings.

(18) Starting from a first embodiment, as illustrated in FIG. 16, a control room (not illustrated) is provided in a front portion of a fuselage 10 of an aircraft, a dividing wall (not illustrated) is provided behind a cockpit, a cabin 12 is provided behind the dividing wall, and a pressure bulkhead 14 is provided in a rear portion of the fuselage 10.

(19) The cabin 12 includes seat groups 16, a longitudinal aisle 18, a latitudinal aisle 20, and aircraft lavatory units 22.

(20) The longitudinal aisle 18 extends in a front-rear direction of the fuselage between the front portion and the rear portion of the fuselage 10.

(21) Two of the seat groups 16 are provided on the left and right sides of the longitudinal aisle 18.

(22) The seat groups 16 each include a plurality of seat rows 1602 arranged in rows in the front-rear direction of the fuselage 10. The seat rows 1602 each include a plurality of seats facing the front of the fuselage 10 and arrayed in a width direction of the fuselage 10.

(23) A plurality of the latitudinal aisles 20 is provided at intervals in the front-rear direction of the fuselage 10, and the latitudinal aisles 20 extend in the width direction of the fuselage 10 and are provided orthogonal to the longitudinal aisle 18.

(24) Boarding entrances 24 are provided at both ends in the extension direction of each of the latitudinal aisles 20. FIG. 16 illustrates the latitudinal aisle 20 provided in the rear portion of the fuselage 10.

(25) Aircraft lavatory units 22 are provided on both sides of the longitudinal aisle 18 in front of the latitudinal aisle 20, and a plurality of galleys 26 partitioned by a dividing wall is provided behind the latitudinal aisle 20.

(26) Thus, the latitudinal aisle 20 is provided between the aircraft lavatory units 22 and the galleys 26, and the aircraft lavatory units 22 are disposed facing both the longitudinal aisle 18 and the latitudinal aisle 20.

(27) The aircraft lavatory unit 22 is disposed behind the seat row 1602 located at the rearmost end of each of the seat groups 16 on a floor of the fuselage 10.

(28) The aircraft lavatory units 22 are configured in a bilaterally symmetrical shape with respect to the longitudinal aisle 18, and the aircraft lavatory unit 22 disposed on the left side of the longitudinal aisle 18 in FIG. 16 will be described.

(29) As illustrated in FIG. 1, a structural frame 22A of the aircraft lavatory unit 22 includes a bottom wall 2202 placed on the floor, a front wall 2204 facing the longitudinal aisle 18, a first side wall 2206 facing the latitudinal aisle 20, a second side wall 2208 facing the rear of the seat row 1602 located at a rear end, a rear wall 2210 adjacent to a wall portion of the fuselage 10, and a ceiling wall 2212.

(30) As illustrated in FIG. 16, the inside of the structural frame 22A is a lavatory 2224 with a toilet 2220, a sink 2222, and the like.

(31) As illustrated in FIG. 1, the front wall 2204 includes a front wall body 2230 and an entrance 2232 of the aircraft lavatory unit 22 (lavatory 2224) provided lateral to the front wall body 2230.

(32) The entrance 2232 includes an opening 2234 provided lateral to the front wall body 2230 and a door 2236 for opening and closing the opening 2234, and the door 2236 is provided with a handle 2238 for opening and closing operation.

(33) An edge portion of the first side wall 2206 on the boarding entrance 24 side is provided in a

curved shape corresponding to the rear wall **2210**, a protruding wall **2240** is provided at the first side wall **2206** on the longitudinal aisle **18** side, and an attendant seat **2242** for seating a cabin attendant is provided on the first side wall **2206** by using the protruding wall **2240**.

(34) The attendant seat **2242** is for two persons, and includes frames **2244** attached to the first side wall **2206**, a backrest **2246** attached to the frames **2244**, and a seat portion **2248** foldably attached to the frames **2244**.

(35) A pump-type bottle **30** that can discharge a disinfectant is provided at each first side wall **2206** of the aircraft lavatory unit **22** that faces the latitudinal aisle **20** and does not include the entrance **2232**, and the pump-type bottle **30** is provided at a height at which a passenger can push down the head portion **3016** thereof.

(36) Here, the height of the head portion **3016** at which a passenger can push down the head portion corresponds to, for example, a position about 80 cm to 140 cm above the floor.

(37) The pump-type bottle **30** is provided at a location of the first side wall **2206** near the boarding entrance **24**.

(38) In the present embodiment, the pump-type bottle **30** is provided at a location of the first side wall **2206** closer to the boarding entrance **24** than the attendant seat **2242**.

(39) As illustrated in FIG. **2**, the pump-type bottle **30** includes a container **3002** containing a disinfectant for hands and fingers, and a pump dispenser **3004** for discharging the disinfectant in the container **3002**. The disinfectant for hands and fingers may be any disinfectant that is discharged from a nozzle portion **3018** to hands and fingers and widely includes a liquid or gel disinfectant.

(40) The container **3002** is formed of a cylindrical shape having a uniform outer diameter and includes, at the center of an upper surface, a small diameter portion having a cylindrical shape and protruding upward. The small diameter portion includes a male thread formed on an outer circumferential surface thereof.

(41) The pump dispenser **3004** includes: a hollow cap **3010** to be screwed to the small diameter portion; a cylinder **3012** held by the cap **3010**, protruding above the cap **3010**, and extending inside the container **3002**; a tube attached to a bottom wall of the cylinder **3012** with a first one-way valve interposed, extending downward, and having a lower end located near a bottom surface of the container **3002**; a piston that slides in the cylinder **3012** and forms a cylinder chamber between the piston and the bottom wall of the cylinder **3012**; a piston rod **3014** protruding above an upper end of the cylinder **3012** from the piston; a spring constantly biasing the piston upward; the head portion **3016** provided at an upper end of the piston rod **3014**; the nozzle portion **3018** protruding from the head portion **3016**; a disinfectant passage communicating from the piston to the nozzle portion **3018**; and a second one-way valve interposed in the disinfectant passage.

(42) Thus, for example, when a hand is released from the head portion **3016** that has been pushed down, the piston is raised by biasing force of the spring to an upper limit position. The rise of the piston causes the pressure in the cylinder chamber to be negative, the second one-way valve is closed, the first one-way valve is opened, and the disinfectant in the container **3002** is sucked into the cylinder chamber via the tube.

(43) When the head portion **3016** is pushed down, the cylinder chamber is compressed by the piston, the second one-way valve is opened, the first one-way valve is closed, the disinfectant in the cylinder chamber rises through the disinfectant passage, and the disinfectant is discharged from the nozzle portion **3018**.

(44) As such a pump-type bottle **30**, commercially available products having known various structures can be used.

(45) As illustrated in FIGS. **2** and **3**, the pump-type bottle **30** is disposed at the first side wall **2206** via an arrangement instrument **34**.

(46) The arrangement instrument **34** includes a lower support member **36** that is attached to the first side wall **2206** and supports a lower portion of the pump-type bottle **30** and an upper support

member **38** that is attached to the first side wall **2206**, separated from the lower support member **36**, and supports an upper portion of the pump-type bottle **30**.

(47) Specifically, the arrangement instrument **34** includes the lower support member **36** that supports the lower portion of the pump-type bottle **30** while regulating the horizontal movement the container **3002** and the upper support member **38** that regulates the upward movement of the pump-type bottle **30** at the upper portion of the pump-type bottle **30**.

(48) As illustrated in FIGS. **4A-4B**, the lower support member **36** includes a lower bracket **36A** to be attached to the first side wall **2206**.

(49) The lower bracket **36A** is made of an alloy or a fiber reinforced plastic and is preferably made of an aluminum alloy in view of strength, rigidity, lightness, processability, and design. In the present embodiment, the lower bracket **36A** is made of an aluminum alloy.

(50) The lower bracket **36A** includes a base plate portion **3602**, an upper plate portion **3604**, a placement plate portion **3606**, and a connecting plate portion **3608**.

(51) The base plate portion **3602** has a rectangular shape and includes first bolt insertion holes **H1** at two locations, both sides of an upper portion of the base plate portion **3602** and at one location, the center of a lower portion of the base plate portion **3602**.

(52) As illustrated in FIG. **3**, the base plate portion **3602** is placed on the first side wall **2206** and is attached to the first side wall **2206** via first bolts **B1** which are male thread members inserted into the first bolt insertion holes **H1**. In the present embodiment, hexagonal socket bolts to be turned by a hexagonal wrench are used as the first bolts **B1**.

(53) As illustrated in FIGS. **4A-4B**, the upper plate portion **3604** protrudes from the upper end of the base plate portion **3602**.

(54) The upper plate portion **3604** includes a lower insertion hole **3604A** having a circular shape through which a lower portion of the container **3002** of the pump-type bottle **30** is removably inserted and which regulates the horizontal movement of the lower portion of the container **3002** with the lower portion of the container **3002** inserted.

(55) The placement plate portion **3606** is protruded from a portion of the base plate portion **3602** located below the upper plate portion **3604**. A bottom portion of the container **3002** inserted through the lower insertion hole **3604A** is placed on the placement plate portion **3606**. The placement plate portion **3606** supports the weight of the pump-type bottle **30** receives the load of the pump-type bottle **30** applied when a passenger or a cabin attendant pushes down the head portion **3016**.

(56) The connecting plate portion **3608** is disposed protruding obliquely upward from a lower end of the base plate portion **3602** and connects ends of the upper plate portion **3604** and the placement plate portion **3606**.

(57) The lower bracket **36A** is provided with strength and rigidity so as to be unbreakable with force of 136 kg-force (kgf) in the vertical direction or force of 136 kg-force (kgf) in the horizontal direction so that the lower bracket **36A** is not broken when a passenger staggers and lays his/her hand on or leans against the pump-type bottle **30** in the case where the aircraft runs into air turbulence to cause a large shake of the fuselage **10** in the vertical direction or in the horizontal direction.

(58) That is, in addition to the material having strength and rigidity for constituting the lower bracket **36A**, the lower bracket **36A** consisting of the base plate portion **3602**, the upper plate portion **3604**, and the connecting plate portion **3608** has a substantially triangular frame shape in a side view as illustrated in FIG. **4B**, in other words, has a closed cross-sectional shape, the shape itself has strength and rigidity, and the lower bracket **36A** is provided with strength and rigidity so as to be unbreakable by force of 136 kg-force (kgf) in the vertical direction or force of 136 kg-force (kgf) in the horizontal direction.

(59) In addition, with the base plate portion **3602** attached to the first side wall **2206**, all of the base plate portion **3602**, the upper plate portion **3604**, and the connecting plate portion **3608** extend in

the horizontal direction, in other words, extend in the same direction. Thus, the lower bracket **36A** can be easily manufactured by extrusion molding, which is advantageous in reducing cost.

(60) As illustrated in FIGS. 5A-5B, the upper support member **38** includes an upper bracket **38A** to be attached to the first side wall **2206**.

(61) Similar to the lower bracket **36A**, the upper bracket **38A** is made of an alloy or a fiber reinforced plastic and is preferably made of an aluminum alloy in view of strength, rigidity, lightness, processability, and design. In the present embodiment, the upper bracket **38A** is made of an aluminum alloy.

(62) The upper bracket **38A** includes a mounting plate portion **3802** and a protruding plate portion **3804**.

(63) The mounting plate portion **3802** has an elongated rectangular shape and includes two second bolt insertion holes H2 in the mounting plate portion **3802**.

(64) As illustrated in FIG. 3, the mounting plate portion **3802** is placed on a location of the first side wall **2206** separated upward from the lower bracket **36A** and is attached to the first side wall **2206** with second bolts B2 which are male thread members inserted into the second bolt insertion holes H2. In the present embodiment, hexagonal socket bolts to be turned by a hexagonal wrench are used as the second bolts B2.

(65) As illustrated in FIGS. 5A-5B, the protruding plate portion **3804** is disposed protruding from an upper end of the mounting plate portion **3802**.

(66) The protruding plate portion **3804** includes an upper insertion hole **3804A** through which the upper portion of the pump-type bottle **30** is removably inserted and which regulates the upward movement of the pump-type bottle **30**.

(67) In the present embodiment, as illustrated in FIG. 2, the upper insertion hole **3804A** is formed to have an inside diameter larger than the outer diameter of the cylinder **3012** constituting the upper portion of the pump-type bottle **30** and smaller than the outer diameter of the cap **3010**. Thus, the cap **3010** comes in contact with a portion of the protruding plate portion **3804** around the upper insertion hole **3804A**, thus regulating the upward movement of the pump-type bottle **30**.

(68) Accordingly, the pump-type bottle **30** can be fixedly supported without falling out of the arrangement instrument **34** including the lower bracket **36A** and the upper bracket **38A** when, for example, the aircraft runs into air turbulence to cause a large shake of the airframe **10** in the vertical direction or in the horizontal direction.

(69) As illustrated in FIG. 3, the wall portions of the structural frame **22A** including the first side wall **2206** includes a core **2250** that has a honeycomb structure and is made of a synthetic fibrous material and a surface plate **2252** that is made of a fiber reinforced plastic and attached to both surfaces of the core **2250**.

(70) A reinforcement frame **40** is provided at a location of the first side wall **2206** to which the lower bracket **36A** and the upper bracket **38A** are attached.

(71) As illustrated in FIGS. 1 and 3, the reinforcement frame **40** includes two thin reinforcement plates **4002** having large contours within which both the base plate portion **3602** of the lower bracket **36A** and the mounting plate portion **3802** of the upper bracket **38A** can be accommodated.

(72) The thin reinforcement plates **4002** are made of a light-weight material having strength and rigidity higher than those of the surface plate **2252**. As such a material, a metal material, a glass fiber reinforced plastic, or the like can be employed.

(73) The thin reinforcement plates **4002** are attached to respective recess portions provided in the core **2250** such that the surfaces of the thin reinforcement plates **4002** and the surfaces of the core **2250** at portions where the thin reinforcement plates **4002** are not provided are positioned on the same planes.

(74) A female thread member **42** having a female thread **4202** that can be engaged with the male thread of the first bolt B1 or the second bolt B2 is embedded in locations of the first side wall **2206** corresponding to the first bolt insertion holes H1 of the base plate portion **3602** and the second bolt

insertion holes H2 of the mounting plate portion **3802**.

(75) The female thread member **42** is made of an alloy and includes a large diameter portion **4204** and a small diameter portion **4206** on each side. The small diameter portions **4206** on both sides are respectively inserted into holes of the thin reinforcement plates **4002** and the surface plates **2252**, and an end surface of the large diameter portion **4204** on each side is disposed in contact with the thin reinforcement plate **4002**.

(76) Accordingly, a portion of the first side surface **2206** where the pump-type bottle **30** is disposed is reinforced by the reinforcement frame **40** so that the lower bracket **36A** and the upper bracket **38A** can be fixedly held by the first side wall **2206** via the female thread members **42** when a passenger staggers and lays his/her hand on or leans against the pump-type bottle **30** due to the aircraft running into air turbulence to apply large force to the lower bracket **36A** and the upper bracket **38A**.

(77) In addition, the reinforcement frame **40** is provided inside the first side wall **2206**, does not disfigure the first side wall **2206** of the aircraft lavatory unit **22**, and is advantageous in disposing the pump-type bottle **30** at the aircraft lavatory unit **22** via the arrangement instrument **34**.

(78) When the pump-type bottle **30** is replenished with the disinfectant, the upper bracket **38A** is detached from the first side wall **2206** by removing the second bolts B2, the pump-type bottle **30** is taken out from the lower bracket **36A**, the container **3002** is replenished with the disinfectant, and then the upper bracket **38A** is attached to the first side wall **2206** with the second bolts B2 again.

(79) Alternatively, the lower bracket **36A** is detached from the first side wall **2206** by removing the first bolts B1, the cap **3010** is removed from the container **3002**, the container **3002** is replenished with the disinfectant, then the cap **3010** is attached to the container **3002**, and the lower bracket **36A** is attached to the first side wall **2206** with the first bolts B1.

(80) That is, detaching at least one of the lower support member **36** or the upper support member **38** from the wall portion allows the pump-type bottle **30** to be taken out from the arrangement instrument **34**, and thus a passenger cannot easily take the pump-type bottle **30** out of the arrangement instrument **34**.

(81) According to such a first embodiment described above, the pump-type bottle **30** that can discharge a disinfectant for hands and fingers is disposed at the first side wall **2206** of the aircraft lavatory unit **22** at a height at which a passenger can push down the head portion **3016**.

(82) Accordingly, when using the aircraft lavatory unit **22**, the passenger can push down the head portion **3016** of the pump-type bottle **30** to discharge a disinfectant to his/her hands and fingers and easily disinfect his/her hands and fingers, and thus opens the door **2236** via the handle **2238** with his/her disinfected hands and fingers to enter the lavatory **2224**.

(83) Then, after using the lavatory **2224**, the passenger can push down the head portion **3016** of the pump-type bottle **30** to discharge a disinfectant to his/her hands and fingers and easily disinfect his/her hands and fingers.

(84) Therefore, even if every passenger touches the handle **2238** of the aircraft lavatory unit **22** on a long-distance flight, it is advantageous in preventing the infections with the novel coronavirus to passengers via the handle **2238** of the aircraft lavatory unit **22** and thus is advantageous in preventing the infections with the novel coronavirus in the aircraft cabin.

(85) In addition, since the pump-type bottle **30** that can discharge a disinfectant for hands and fingers is disposed at the first side wall **2206** constituting the latitudinal aisle **20** near the boarding entrance **24**, passengers can disinfect their hands and fingers when boarding the aircraft and then board the aircraft, which is advantageous in preventing the infections with the novel coronavirus in the cabin.

(86) Further, since the pump-type bottle **30** that can discharge a disinfectant for hands and fingers is disposed lateral to the attendant seat **2242** for seating a cabin attendant, when the cabin attendant seated on the attendant seat **2242** receives a call from a passenger and leaves the seat, the cabin attendant can easily disinfect his/her hands and fingers using the pump-type bottle **30** before

coming to the seat of the passenger from whom the call was received. Then, after serving the passenger and before sitting on the attendant seat **2242**, the cabin attendant can easily disinfect his/her hands and fingers using the pump-type bottle **30**.

(87) Thus, it is also advantageous in preventing the infections with the novel coronavirus to passengers via cabin attendants in the cabin.

(88) In addition, the arrangement instrument **34** is used to dispose the pump-type bottle **30** at the first side wall **2206** and the arrangement instrument **34** includes two separate members, i.e., the lower support member **36** for supporting the lower portion of the pump-type bottle **30** and the upper support member **38** for supporting the upper portion of the pump-type bottle **30**, which is advantageous in reducing the weight and cost of the arrangement instrument **34**. Thus, it is possible to dispose the pump-type bottle **30** at the first side wall **2206** while reducing the weight and cost.

(89) Next, a second embodiment will be described with reference to FIG. 6.

(90) Note that, in the embodiments described below, the same sections and members as those of the first embodiment will be denoted by the same reference signs, the descriptions thereof will be simplified or omitted, and different sections from the first embodiment will be mainly described.

(91) The second embodiment differs from the first embodiment in a placement position of the pump-type bottle **30**.

(92) That is, in the second embodiment, the pump-type bottle **30** is disposed at the front wall body **2230** of the front wall **2204** facing the longitudinal aisle **18** via the arrangement instrument **34** at a height at which a passenger can push down the head portion **3016** thereof, that is, the pump-type bottle **30** is disposed lateral to the entrance **2232**.

(93) As illustrated in FIG. 3, similar to the first side wall **2206**, the front wall body **2230** includes the core **2250** that has a honeycomb structure and is made of a synthetic fibrous material and the surface plate **2252** that is made of a fiber reinforced plastic and attached to both surfaces of the core **2250**, and the reinforcement frame **40** including the two thin reinforcement plates **4002** is provided at a portion of the front wall body **2230** to which the lower bracket **36A** and the upper bracket **38A** are attached.

(94) The first bolts **B1** and the second bolts **B2** respectively inserted into the first bolt insertion holes **H1** of the base plate portion **3602** of the lower bracket **36A** and the second bolt insertion holes **H2** of the mounting plate portion **3802** of the upper bracket **38A** are engaged with the female thread members **42** that are embedded in the front wall body **2230** and sandwiched between the thin reinforcement plates **4002**.

(95) According to such a second embodiment described above, the pump-type bottle **30** that can discharge a disinfectant for hands and fingers is disposed at the front wall body **2230** lateral to the entrance **2232** of the aircraft lavatory unit **22** at a height at which a passenger can push down the head portion **3016**.

(96) Accordingly, when using the aircraft lavatory unit **22**, the passenger can push down the head of the pump-type bottle **30** lateral to the entrance **2232** to discharge a disinfectant to his/her hands and fingers and easily disinfect his/her hands and fingers, and thus opens the door **2236** via the handle **2238** with his/her disinfected hands and fingers to enter the lavatory **2224**.

(97) Then, after using the lavatory **2224**, the passenger can push down the head of the pump-type bottle **30** lateral to the entrance **2232** to discharge a disinfectant to his/her hands and fingers and easily disinfect his/her hands and fingers.

(98) That is, since the pump-type bottle **30** is located lateral to the entrance **2232**, passengers can disinfect their hands and fingers before and after using the aircraft lavatory unit **22**.

(99) Therefore, even if every passenger touches the handle **2238** of the aircraft lavatory unit **22** on a long-distance flight, it is advantageous in preventing the infections with the novel coronavirus to passengers via the handle **2238** of the aircraft lavatory unit **22** and thus is advantageous in preventing the infections with the novel coronavirus in the aircraft cabin.

(100) Further, since the pump-type bottle **30** that can discharge a disinfectant for hands and fingers

is disposed near the attendant seat **2242**, when a cabin attendant seated on the attendant seat **2242** receives a call from a passenger and leaves the seat, the cabin attendant can easily disinfect his/her hands and fingers using the pump-type bottle **30** before coming to the seat of the passenger from whom the call was received. Then, after serving the passenger and before sitting on the attendant seat **2242**, the cabin attendant can easily disinfect his/her hands and fingers using the pump-type bottle **30**.

(101) Thus, it is also advantageous in preventing the infections with the novel coronavirus to passengers via cabin attendants in the cabin.

(102) In addition, the arrangement instrument **34** is used to dispose the pump-type bottle **30** at the front wall body **2230** and the arrangement instrument **34** includes two separate members, i.e., the lower support member **36** for supporting the lower portion of the pump-type bottle **30** and the upper support member **38** for supporting the upper portion of the pump-type bottle **30**, which is advantageous in reducing the weight and cost of the arrangement instrument **34**. Thus, it is possible to dispose the pump-type bottle **30** at the front wall body **2230** while reducing the weight and cost.

(103) Next, a third embodiment will be described with reference to FIGS. 7A-7B.

(104) The third embodiment has a different configuration from that of the lower bracket **36A** of the first and the second embodiments and the other configurations similar to that of the first and the second embodiments.

(105) First, in a lower bracket **36B** of the third embodiment, a lower insertion hole **3604B** provided in the upper plate portion **3604** is formed in a rectangular shape in the third embodiment, while the lower insertion hole is formed in a circular shape in the first and the second embodiments.

(106) Similar to the first embodiment, the lower portion of the container **3002** of the pump-type bottle **30** is removably inserted into the lower insertion hole **3604B** having a rectangular shape, and the lower insertion hole **3604B** regulates the horizontal movement of the lower portion of the container **3002** with the lower portion of the container **3002** inserted therein.

(107) The lower bracket **36B** includes a reinforcement plate portion **3610** in addition to the base plate portion **3602**, the upper plate portion **3604**, the placement plate portion **3606**, and the connecting plate portion **3608**.

(108) The reinforcement plate portion **3610** protrudes obliquely downward from the base plate portion **3602** at a portion from which the placement plate portion **3606** protrudes and is connected to the connecting plate portion **3608**.

(109) Similar to the first and the second embodiments, as illustrated in FIG. 7B, the lower bracket **36B** has a substantially triangular frame shape in a side view formed by the base plate portion **3602**, the upper plate portion **3604**, and the connecting plate portion **3608**, in other words, has a closed cross-sectional shape, and thus the shape itself has strength and rigidity. In addition, in the third embodiment, since the reinforcement plate portion **3610** is provided, the strength and the rigidity of the lower bracket **36B** are further enhanced.

(110) Specifically, the lower bracket **36B** is provided with strength and rigidity so as to be unbreakable with force of 136 kg-force (kgf) in the vertical direction or force of 136 kg-force (kgf) in the horizontal direction so that the lower bracket **36B** is not broken when a passenger staggers and lays his/her hand on or leans against the pump-type bottle **30** in the case where the aircraft runs into air turbulence.

(111) In the lower bracket **36B** of the third embodiment, similar to the first and the second embodiment, with the base plate portion **3602** attached to the first side wall **2206** or the front wall body **2230**, all of the base plate portion **3602**, the upper plate portion **3604**, the connecting plate portion **3608**, and the reinforcement plate portion **3610** extend in the horizontal direction, in other words, extend in the same direction. Thus, the lower bracket **36B** can be easily manufactured by extrusion molding, which is advantageous in reducing cost.

(112) The third embodiment differs from the first embodiment only in the structure of the lower bracket **36B**, and thus the same effects as in the first embodiment are also exhibited by the third

embodiment, which is advantageous in preventing the infections with the novel coronavirus in the aircraft cabin.

(113) Next, a fourth embodiment will be described with reference to FIGS. **8** to **10B**.

(114) The fourth embodiment has a different configuration from that of the lower bracket **36A** of the first embodiment and the other configurations similar to that of the first embodiment.

(115) Note that, in the following embodiments, cases where the pump-type bottle **30** is disposed at the first side wall **2206** will be described. However, as a matter of course, the following embodiments can be applied to cases where the pump-type bottle **30** is disposed at the front wall body **2230**.

(116) As illustrated in FIGS. **8** and **9**, a lower bracket **36C** of the fourth embodiment is attached to the first side wall **2206**. Similar to the first embodiment, the lower bracket **36C** is made of an alloy or a fiber reinforced plastic and is preferably made of an aluminum alloy in view of strength, lightness, processability, and design. In the present embodiment, the lower bracket **36C** is made of an aluminum alloy.

(117) As illustrated in FIGS. **10A-10B**, the lower bracket **36C** includes a peripheral wall **3620** and a bottom wall **3630**.

(118) The peripheral wall **3620** is a portion into which the lower portion of the container **3002** of the pump-type bottle **30** is inserted and which regulates the horizontal movement of the pump-type bottle **30**.

(119) The peripheral wall **3620** includes a base plate portion **3622** to be placed on the first side wall **2206** and attached to the first side wall **2206** and having an elongated shape, a side plate portion **3624** on each side that protrudes from each side of the base plate portion **3622**, and a curved plate portion **3626** that connects ends of the side plate portions **3624** on both sides.

(120) An inner surface of the base plate portion **3622**, inner surfaces of the side plate portions **3624** on both sides, and an inner surface of the curved plate portion **3626** are formed in such dimensions that the lower portion of the container **3002** of the pump-type bottle **30** can be inserted and removed and that the horizontal movement of the lower portion of the container **3002** can be regulated.

(121) A third bolt insertion hole **H3** is provided on each side of an upper portion of the base plate portion **3622**, and as illustrated in FIG. **9**, male threads of third bolts **B3** (hexagonal socket bolts), which are male thread members inserted into the third bolt insertion holes **H3**, are engaged with the female threads **4202** of the female thread members **42** embedded in locations of the first side wall **2206** reinforced by the reinforcement frame **40**, similar to the first embodiment.

(122) The bottom wall **3630** is a portion that is provided at the peripheral wall **3620** and supports the lower portion of the pump-type bottle **30**.

(123) As illustrated in FIGS. **10A-10B**, the bottom wall **3630** includes a plurality of wall portions that connect lower portions of the inner surfaces of the peripheral wall **3620** facing each other inside the peripheral wall **3620**.

(124) In the present embodiment, the bottom wall **3630** includes a first bottom wall portion **3632** and a second bottom wall portion **3634** that are orthogonal to each other.

(125) The first bottom wall portion **3632** connects a central portion in the extension direction of the curved plate portion **3626** and a central portion in the extension direction of the base plate portion **3622**.

(126) The second bottom wall portion **3634** connects the boundary portions between the curved plate portion **3626** and the side plate portions **3624** mutually opposed.

(127) The lower bracket **36C** consisting of the peripheral wall **3620** and the bottom wall **3630** has a substantially rectangular frame shape including the curved plate portion **3626** in a plan view, in other words, has a closed cross-sectional shape, includes the bottom wall **3630** that connects mutually-opposed portions of the inner surfaces of the peripheral wall **3620**, and thus the shape itself has strength and rigidity.

(128) In addition, with the base plate portion **3622** attached to the first side wall **2206**, the peripheral wall **3620** and the bottom wall **3630** extend in the vertical direction, in other words, extend in the same direction. Thus, the lower bracket **36C** can be easily manufactured by extrusion molding, which is advantageous in reducing cost.

(129) The lower bracket **36C** is provided by cutting the first bottom wall portion **3632** and the second bottom wall portion **3634** to heights lower than the height of the peripheral wall **3620** after the extrusion molding.

(130) The fourth embodiment differs from the first embodiment only in the structure of the lower bracket **36C**, and thus the same effects as in the first embodiment are also exhibited by the fourth embodiment, which is advantageous in preventing the infections with the novel coronavirus in the aircraft cabin.

(131) Next, a fifth embodiment will be described with reference to FIGS. **11A-11B**.

(132) The fifth embodiment is a modified example of the fourth embodiment, and differs from the fourth embodiment in the structure of the bottom portion of the lower bracket **36C**.

(133) That is, while the bottom portion of the lower bracket **36C** of the fourth embodiment includes the first bottom wall portion **3632** and the second bottom wall portion **3634**, the bottom portion of a lower bracket **36D** of the fifth embodiment includes two bolts, a fourth bolt **B4** (hexagonal socket bolt) and a fifth bolt **B5** (hexagonal socket bolt).

(134) Similar to the fourth embodiment, the peripheral wall **3620** includes the base plate portion **3622** having an elongated shape, the side plate portions **3624** on both sides, and the curved plate portion **3626**, and the third bolt insertion hole **H3** is provided on each side of the upper portion of the base plate portion **3622**.

(135) Further, as illustrated in FIG. **11B**, a fourth bolt insertion hole **H4** is coaxially provided at each of the center of a lower portion of the base plate portion **3622** and the center of a lower portion of the curved plate portion **3626** facing the center of the lower portion of the base plate portion **3622**.

(136) Fifth bolt insertion holes **H5** are coaxially provided at portions of the side plate portions **3624** on both sides close to the curved plate portion **3626**.

(137) The third bolt **B3** is inserted into each of the third bolt insertion holes **H3** in the upper portion of the base plate portion **3622**, and similar to the fourth embodiment illustrated in FIG. **9**, the male threads of the third bolts **B3** are engaged with the female threads **4202** of the female thread members **42** embedded in the locations of the first side wall **2206** reinforced by the reinforcement frame **40**.

(138) As illustrated in FIG. **11B**, the fourth bolt **B4** (hexagonal socket bolt) is inserted from the fourth bolt insertion hole **H4** provided in the curved plate portion **3626** to the fourth bolt insertion hole **H4** provided in the base plate portion **3622**, and a male thread of the fourth bolt **B4** is engaged with the female thread **4202** of the female thread member **42** embedded in the first side wall **2206** at the portion reinforced by the reinforcement frame **40**.

(139) Thus, the attachment strength of the lower bracket **36D** is increased as compared with the fourth embodiment.

(140) The fifth bolt **B5** is inserted from one of the fifth bolt insertion holes **H5** to the other one of the fifth bolt insertion holes **H5**, and a nut **N** is engaged with a male thread portion of the fifth bolt **B5** protruding to the outside of the side plate portion **3624**, thereby the fifth bolt **B5** is attached to the side plate portions **3624** on both sides.

(141) The fourth bolt **B4** and the fifth bolt **B5** are orthogonal to each other in a plan view, and the fifth bolt **B5** is located above the fourth bolt **B4**.

(142) With the base plate portion **3622** attached to the first side wall **2206**, the base plate portion **3622**, the side plate portions **3624** on both sides, and the curved plate portion **3626** extend in the vertical direction, and thus the peripheral wall **3620** can be easily manufactured by extrusion molding, which is advantageous in reducing cost.

(143) In removing the lower bracket **36D** from the first side wall **2206**, the fourth bolt **B4** in addition to the two third bolts **B3** is removed from the female thread members **42** of the first side wall **2206**.

(144) The fifth embodiment differs from the first embodiment only in the structure of the lower bracket **36D**, and thus the same effects as in the first embodiment are also exhibited by the fifth embodiment, which is advantageous in preventing the infections with the novel coronavirus in the aircraft cabin.

(145) Next, a sixth embodiment will be described with reference to FIGS. **12** and **13**.

(146) The sixth embodiment has a different configuration from that of the upper bracket **38A** of the first embodiment and the other configurations similar to that of the first embodiment.

(147) Similar to the first embodiment, an upper bracket **38B** of the sixth embodiment is made of an alloy or a fiber reinforced plastic and is preferably made of an aluminum alloy in view of strength, lightness, processability, and design. In the present embodiment, the upper bracket **38B** is made of an aluminum alloy.

(148) The upper bracket **38B** includes a mounting plate portion **3810** and a clamping portion **3820**, and the mounting plate portion **3810** has an elongated rectangular shape, and two sixth bolt insertion holes **H6** are provided in the mounting plate portion **3810**.

(149) As illustrated in FIG. **13**, the mounting plate portion **3810** is placed on a location of the first side wall **2206** separated upward from the lower bracket **36A** and is attached to the first side wall **2206** by using sixth bolts **B6** (hexagonal socket bolts) which are male thread members inserted into the sixth bolt insertion holes **H6**.

(150) Similar to the first embodiment, the male threads of the sixth bolts **B6** are engaged with the female threads **4202** of the female thread members **42** embedded in the locations of the first side wall **2206** reinforced by the reinforcement frame **40**.

(151) A pair of arm portions **3812** are disposed protruding from the center in the length direction of the mounting plate portion **3810**, and the clamping portions **3820** are provided at tip portions of the arm portions **3812**.

(152) The clamping portion **3820** includes two clamping pieces **3822** having an arcuate shape facing each other.

(153) In the present embodiment, base portions of the two clamping pieces **3822** are connected.

(154) The two clamping pieces **3822** are configured to be able to clamp the cap **3010** at the upper portion of the pump-type bottle **30**.

(155) Thus, by pushing the cap **3010** into between the two clamping pieces **3822**, the cap **3010** is clamped by the two clamping pieces **3822**, and the horizontal movement and the upward movement of the cap **3010** are regulated by the two clamping pieces **3822**.

(156) Similar to the first embodiment, the lower portion of the pump-type bottle **30** is supported from below, and the horizontal movement thereof is regulated by the lower bracket **36A**.

(157) For example, when the pump-type bottle **30** is replenished with a disinfectant, the upper bracket **38B** is removed from the first side wall **2206** by removing the two sixth bolts **B6** from the female thread members **42** of the first side wall **2206**.

(158) The sixth embodiment differs from the first embodiment only in the structure of the upper bracket **38B**, and thus the same effects as in the first embodiment are also exhibited by the sixth embodiment, which is advantageous in preventing the infections with the novel coronavirus in the aircraft cabin.

(159) Next, a seventh embodiment will be described with reference to FIGS. **14** and **15**.

(160) The seventh embodiment has a different configuration of the upper support member **38** of the arrangement instrument **34** from that of the first embodiment in the configuration and the other configurations similar to that of the first embodiment.

(161) In the seventh embodiment, the upper support member **38** includes a band **38C** having an elongated shape that is made of an elastic material and has flexibility. The band **38C** is embedded

with a mesh made of a metal so that the band **38C** is not easily broken and a passenger cannot easily take the pump-type bottle **30** out of the arrangement instrument **34**.

(162) Seventh bolt insertion holes **H7** are provided at both ends in the longitudinal direction of the band **38C**, and the band **38C** is attached to a location of the first side wall **2206** separated upward from the lower bracket **36A** by using seventh bolts **B7** (hexagonal socket bolts) which are male thread members inserted into the seventh bolt insertion holes **H7** as illustrated in FIG. **15**.

(163) Similar to the first embodiment, the male threads of the seventh bolts **B7** are engaged with the female threads **4202** of the female thread members **42** embedded in the locations of the first side wall **2206** reinforced by the reinforcement frame **40**.

(164) An intermediate portion of the band **38C** in the longitudinal direction comes in elastic contact with the cap **3010** at the upper portion of the pump-type bottle **30** to bias the cap **3010** toward the first side wall **2206** and regulates the horizontal movement and the upward movement of the upper portion of the pump-type bottle **30**.

(165) In addition, the lower portion of the pump-type bottle **30** is supported from below, and the horizontal movement thereof is regulated by the lower bracket **36A**.

(166) In taking out the pump-type bottle **30** from the arrangement instrument **34**, such as in replenishing a disinfectant or the like, the seventh bolt **B7** located at one end of the band **38C** in the longitudinal direction is removed from the first side wall **2206**. This requires a troublesome operation of turning the seventh bolt **B7**, and thus a passenger cannot easily take the pump-type bottle **30** out of the arrangement instrument **34**.

(167) The seventh embodiment differs from the first embodiment only in the structure of the upper support member **38**, and thus the same effects as in the first embodiment are also exhibited by the seventh embodiment, which is advantageous in preventing the infections with the novel coronavirus in the aircraft cabin.

(168) Note that the configurations of the upper support member **38** of the sixth and the seventh embodiments can also be applied to the first to the fifth embodiments as a matter of course.

Claims

1. An aircraft lavatory unit disposed in an aircraft comprising: a longitudinal aisle extending in a front-rear direction of a fuselage, a latitudinal aisle extending perpendicularly to the longitudinal aisle, and seat groups provided on both sides of the longitudinal aisle, the aircraft lavatory unit being configured to face at least one of the longitudinal aisle or the latitudinal aisle, the aircraft lavatory unit comprising: a structural frame comprising a wall portion facing the at least one of the longitudinal aisle or the latitudinal aisle; and, a pump-type bottle that can discharge a disinfectant for hands and fingers being disposed at the wall portion via an arrangement instrument at a height at which a passenger can push down a head portion of the pump-type bottle, wherein the arrangement instrument includes a lower support member that supports a lower portion of the pump-type bottle while restricting movement of the pump-type bottle in a horizontal direction, and an upper support member that is separated from the lower support member and restricts upward movement of the pump-type bottle in an upper portion of the pump-type bottle, each of the lower support member and the upper support member is detachably directly attached to the wall portion by a male screw member, and removal of the pump-type bottle from the arrangement instrument is enabled only by removal of at least one of the lower support member or the upper support member from the wall.

2. The aircraft lavatory unit according to claim 1, wherein: the structural frame comprises a side wall provided facing the latitudinal aisle; and, the wall portion is the side wall.

3. The aircraft lavatory unit according to claim 2, wherein: a boarding entrance is disposed at an end portion in an extension direction of the latitudinal aisle; and, the pump-type bottle is disposed at a location of the side wall close to the boarding entrance.

4. The aircraft lavatory unit according to claim 2, wherein: an attendant seat for seating a cabin attendant is disposed at the side wall on a side of the longitudinal aisle; and, the pump-type bottle is disposed at a location of the side wall closer to the boarding entrance than the attendant seat.
 5. The aircraft lavatory unit according to claim 1, wherein: the structural frame comprises a front wall comprising a front wall body provided facing the longitudinal aisle and an entrance provided lateral to the front wall body; and, the wall portion is the front wall body.
 6. The aircraft lavatory unit according to claim 5, wherein a location of the wall portion to which the arrangement instrument is attached is reinforced by a reinforcement frame.
 7. The aircraft lavatory unit according to claim 1, wherein a location of the wall portion to which the arrangement instrument is attached is reinforced by a reinforcement frame.
 8. The aircraft lavatory unit according to claim 1, wherein the removal of the pump-type bottle from the arrangement instrument is enabled by the removal of the lower support member from the wall.
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